



NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY
K H U L N A

***UI/UX Design Documentation and Development Planning for AI Prompt
Engineering Learning Hub***

Course Code: CSE4104

Course Title: Software Development III

Team: CSE4104-7C-T02

Department of Computer Science and Engineering

Northern University of Business and Technology, Khulna

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Team Information Sheet

Team Name:	CSE4104-7C-T02
Section:	7C
Project Title:	AI Prompt Engineering Learning Hub
GitHub Repository Link:	https://github.com/RahminHossain/cse4104-7c-t02-ai-prompt-engineering-learning-hub
Team Leader:	Rahmin Hossain

Team Members Roles & Credentials

Sl	Members Name	Student ID	Role/Responsibility
1	Lamiha Faija	11230121140	Frontend Developer & UI Researcher
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3	Masum Billah	11230121149	Backend Developer & QA Tester
4	Rahmin Hossain	11230121156	Database Admin & Data Engineer

1. Project Information

Project Title: AI Prompt Engineering Learning Hub

1.1 Project Overview

The AI Prompt Engineering Learning Hub is a web-based educational platform designed to help users learn and practice prompt engineering for generative AI tools. It provides an interactive environment where users can write prompts, test them, and receive real-time feedback from an AI system.

1.2 Problem Statement

Many users struggle to write effective prompts for AI systems. Current learning resources like blogs and videos are static and do not provide hands-on practice or feedback. As a result:

1. Users rely on trial and error.
2. Outputs are often low-quality.
3. There is no centralized platform for structured learning.

This project aims to solve these issues by creating an interactive and intelligent learning system.

1.3 Objectives

1. Develop an interactive platform for learning prompt engineering
2. Provide automated AI-based prompt evaluation and feedback
3. Create sandbox environments for testing prompts
4. Introduce gamification (XP, badges, challenges) to improve engagement

2. UI / Wireframe Design

2.1 Landing/Home Page

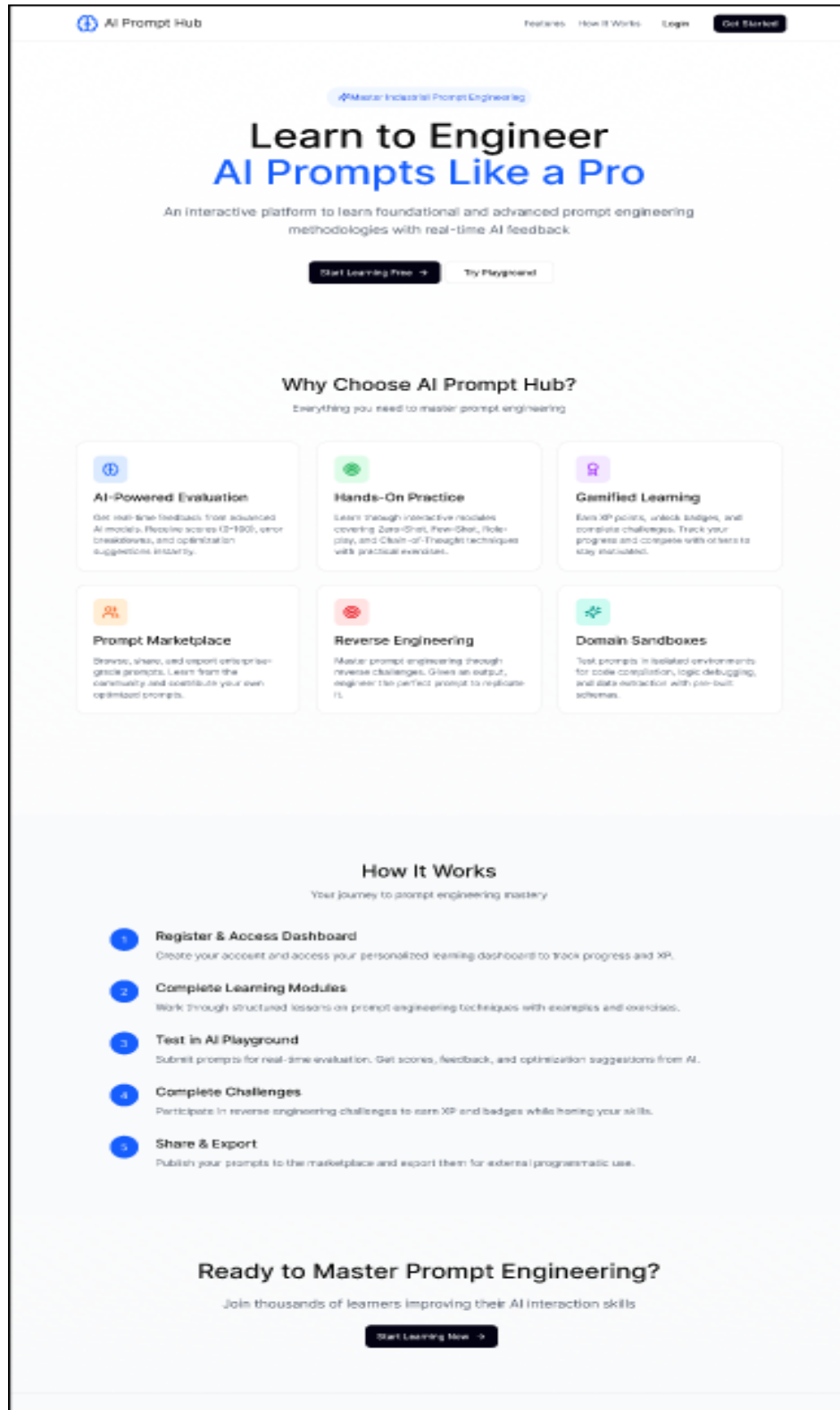


Fig. 1: Landing / Home Page

This is the primary landing and home page of the platform designed to introduce visitors to the core concepts of prompt engineering. It features a prominent hero section titled "Learn to Engineer AI Prompts Like a Pro," complemented by call-to-action buttons for seamless onboarding. The page layout breaks down the unique value propositions under "Why Choose AI Prompt Hub?", provides a structured step-by-step visual workflow under "How It Works" to guide new users, and includes navigation anchors for registration or login channels.

2.2 Login page

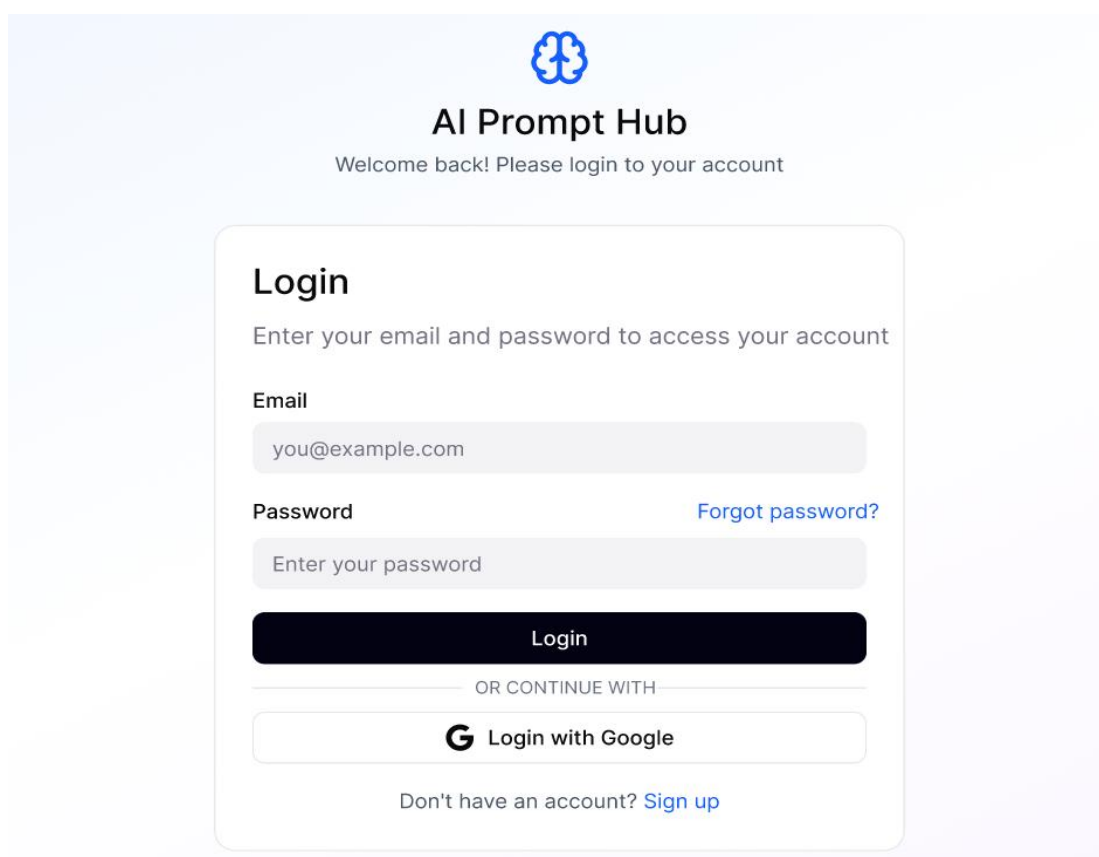
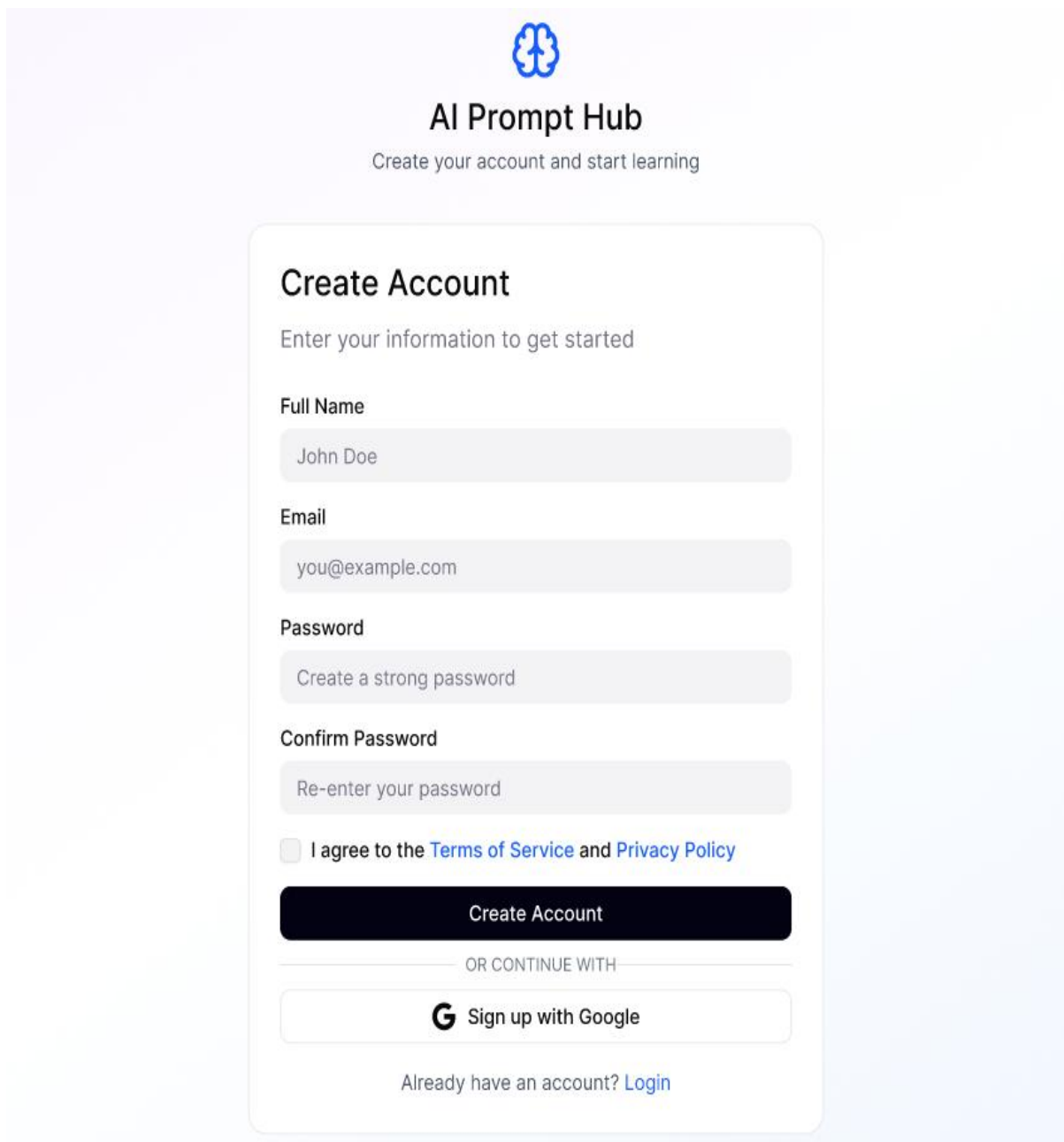
The image shows a login page for "AI Prompt Hub". At the top, there is a blue brain icon. Below it, the text "AI Prompt Hub" is centered, followed by "Welcome back! Please login to your account". The main login form is a white box with a rounded border. It has a "Login" heading and a subtext "Enter your email and password to access your account". There are two input fields: "Email" with the placeholder "you@example.com" and "Password" with the placeholder "Enter your password". A "Forgot password?" link is next to the password field. Below the fields is a dark blue "Login" button. Underneath the button is a horizontal line with the text "OR CONTINUE WITH". Below this is a "Login with Google" button featuring the Google logo. At the bottom of the form, there is a link "Don't have an account? Sign up".

Fig. 2: Login Page

This page showcases the secure authentication interface for returning users to access their personalized profiles and dashboards. It includes explicit text input fields for "Email" and "Password" verification along with a "Forgot password?" link for account recovery. Additionally, it provides alternative one-click "Login with Google" social authentication integration and an onboarding redirect link labeled "Sign up" for users who do not possess a registered account yet.

2.3 Registration Page



The registration page for AI Prompt Hub features a light purple header with a blue brain icon and the text "AI Prompt Hub" and "Create your account and start learning". The main content is a white rounded rectangle with the title "Create Account" and the instruction "Enter your information to get started". It contains five input fields: "Full Name" (with "John Doe"), "Email" (with "you@example.com"), "Password" (with "Create a strong password"), "Confirm Password" (with "Re-enter your password"), and a checkbox for "I agree to the Terms of Service and Privacy Policy". Below these is a dark blue "Create Account" button, followed by a horizontal line and the text "OR CONTINUE WITH". Below this is a white button with the Google logo and "Sign up with Google". At the bottom, it says "Already have an account? Login" with a blue link.

AI Prompt Hub

Create your account and start learning

Create Account

Enter your information to get started

Full Name

John Doe

Email

you@example.com

Password

Create a strong password

Confirm Password

Re-enter your password

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

Create Account

OR CONTINUE WITH

 Sign up with Google

Already have an account? [Login](#)

Fig. 3: Registration Page

This page outlines the platform's registration interface, allowing new visitors to create an account by filling out their Full Name, Email, Password, and Confirm Password fields. It enforces user acknowledgment via a Terms of Service and Privacy Policy checkbox prior to profile generation. It also features an integrated social registration pathway through a "Sign up with Google" option and includes a bottom anchor redirecting existing account holders back to the Login page.

2.4 Dashboard

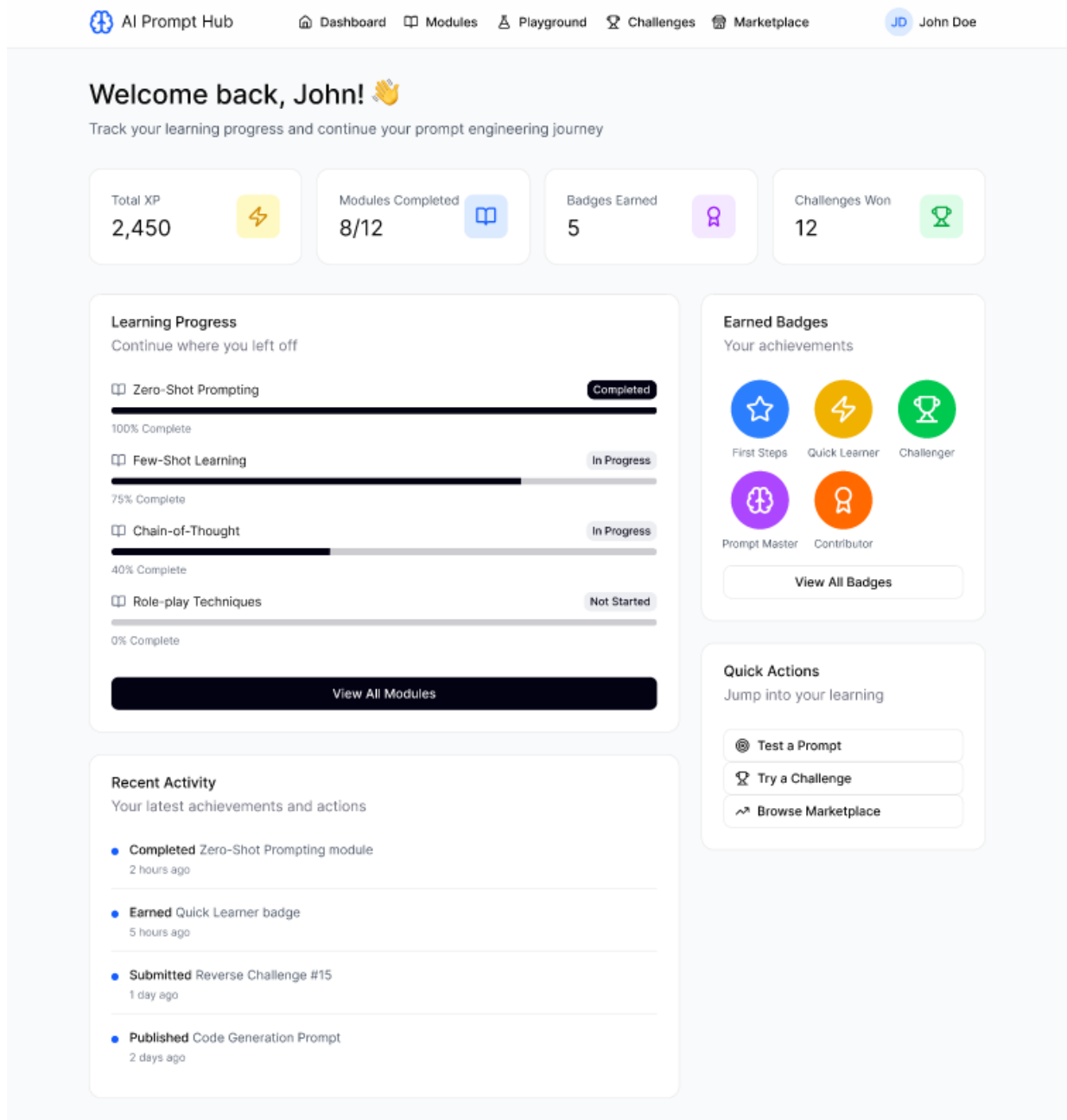


Fig. 4: Dashboard

This is the personalized user dashboard that greets the user upon successful login. It tracks the user's total learning progress, displays overall stats (Total XP, Modules Completed, Badges Earned, and Challenges Won), visualizes current course progress (e.g., Zero-Shot Prompting, Few-Shot Learning), lists unlocked badges, and provides quick action links alongside a recent activity log to ensure a continuous learning experience.

2.5 User Profile Page

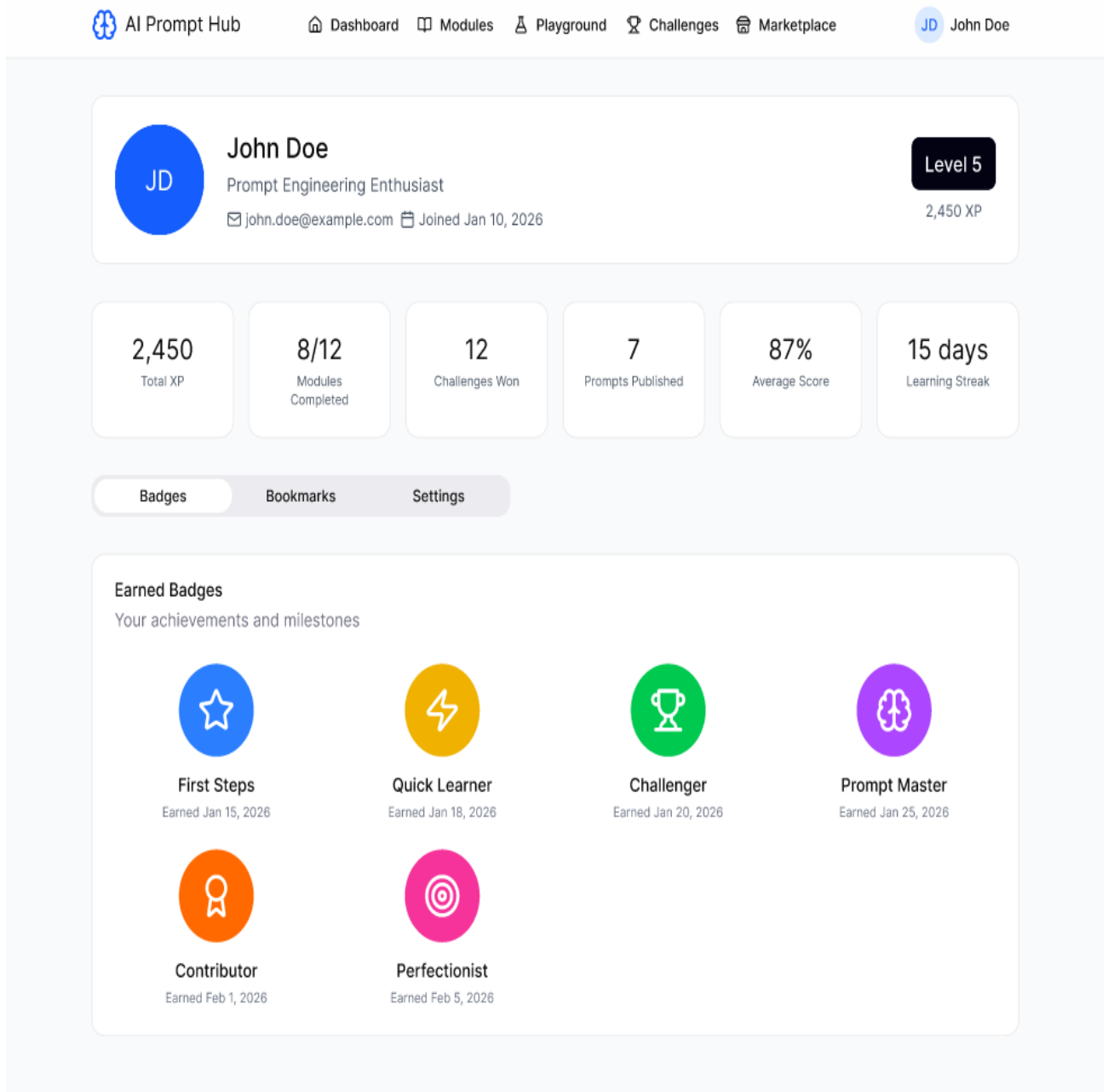


Fig. 5: User Profile Page

This page displays the detailed profile of the authenticated user. It highlights personal credentials, user level, and comprehensive performance metrics such as Total XP, Prompts Published, Average Score, and Learning Streak. It also features dedicated sub-tabs for viewing all "Earned Badges" with their unlock dates, managed "Bookmarks," and account "Settings."

Bookmarks

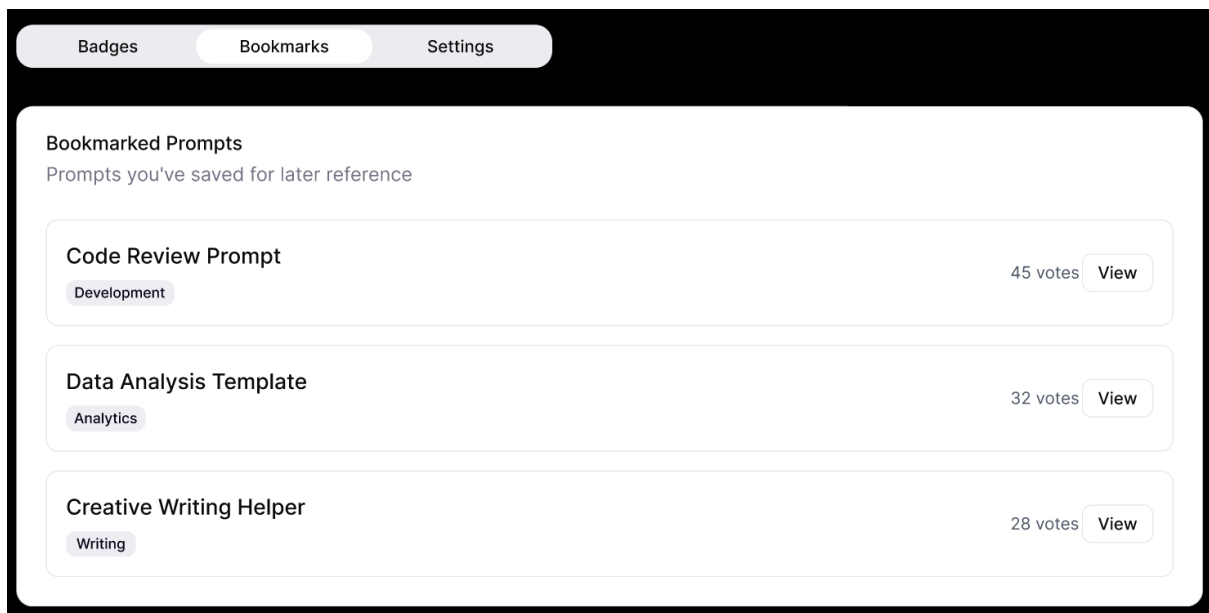


Fig. 6: Bookmarks

This sub-view displays the user's bookmarked items within their profile. It acts as a personalized library titled "Bookmarked Prompts," where users can easily access and reference saved prompt templates categorized under distinct domains like Development, Analytics, and Writing.

Settings

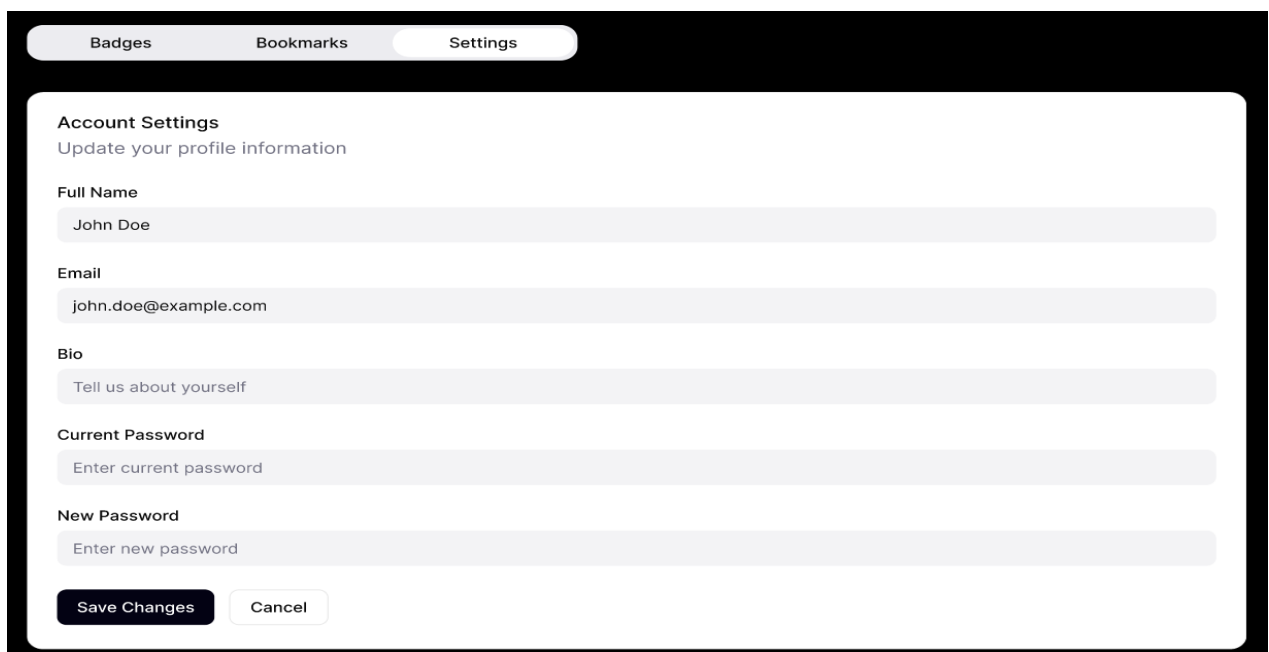


Fig. 7: Settings

This sub-view illustrates the "Account Settings" form accessible from the user's profile tabs. It allows authenticated users to seamlessly update their profile configurations, including changing their Full Name, Email address, Bio, and securely modifying their account credentials via the "Current Password" and "New Password" fields.

2.6 Main Functional Page

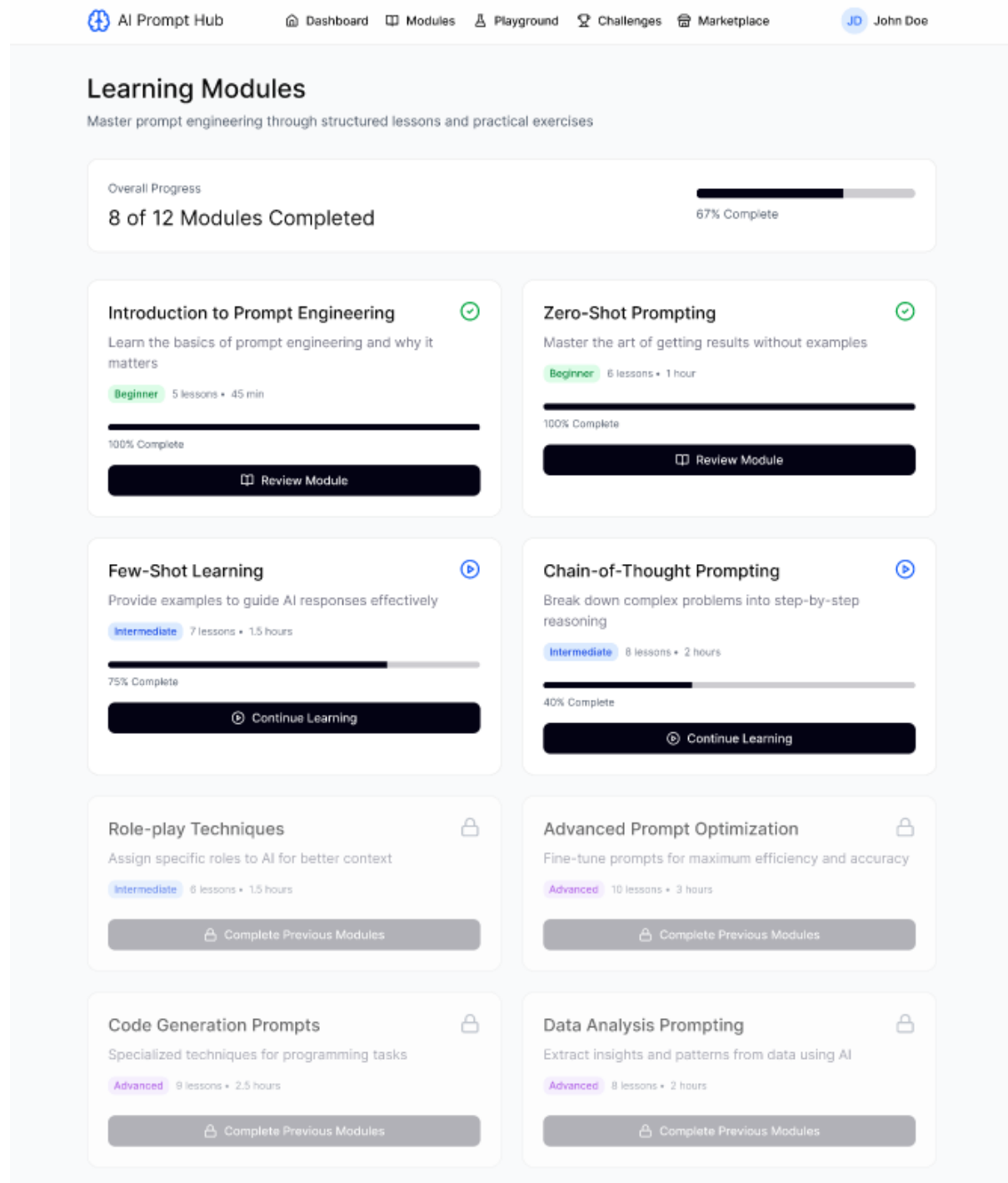


Fig. 8: Main Functional Page

This page acts as the central hub for structured learning on the platform, titled "Learning Modules." It lists various course levels ranging from beginner concepts (such as Introduction to Prompt Engineering and Zero-Shot Prompting) to intermediate and advanced strategies. Users can view their overall module completion progress, track individual module statuses, and directly jump into a section by clicking "Continue Learning" or "Review Module."

Module Details

The screenshot displays the 'Few-Shot Learning' module page within the 'AI Prompt Hub'. The top navigation bar includes links to Dashboard, Modules, Playground, Challenges, and Marketplace, along with a user profile for John Doe. The module page is titled 'Intermediate Few-Shot Learning' and describes the goal: 'Provide examples to guide AI responses effectively'. It shows a progress bar for 7 lessons, with 5 completed (75%).

What You'll Learn

- Understand few-shot learning principles
- Select effective examples
- Structure prompts with examples
- Apply few-shot techniques in practice
- Optimize example selection

Lessons
Complete all lessons to finish this module

Lesson Title	Duration	Status	Action
What is Few-Shot Learning?	8 min	Completed	Review
Providing Examples	12 min	Completed	Review
Example Selection Strategies	15 min	Completed	Review
Practical Exercise: Few-Shot Classification	20 min	Completed	Review
Common Mistakes to Avoid	10 min	Completed	Review
Advanced Techniques	18 min	Not Started	Start
Final Quiz	15 min	Not Started	Start

Example Prompts
Learn from these practical examples

- Task Classification**
Classify user requests into categories
- Sentiment Analysis**
Analyze sentiment with few-shot examples

Module Resources

- Download Cheat Sheet
- Watch Video Tutorial
- Test in Playground

Earn Rewards

- Complete all lessons: +200 XP
- Pass final quiz: +100 XP
- Perfect score: Badge

Fig. 9: Module Details

This sub-view displays the inner interface of a specific learning module, such as "Few-Shot Learning." It shows the detailed curriculum breakdown consisting of sequential lessons (e.g., What is Few-Shot Learning?, Providing Examples) and a Final Quiz. The interface tracks step-by-step progress, displays learning outcomes ("What You'll Learn"), provides downloadable cheat sheets and videos under "Module Resources," and highlights gamified rewards such as XP and badges for successful completion.

Challenges

The screenshot displays the 'AI Prompt Hub' interface, specifically the 'Challenges' section. The top navigation bar includes links to 'Dashboard', 'Modules', 'Playground', 'Challenges', and 'Marketplace', along with a user profile 'JD John Doe'. The main heading is 'Reverse Engineering Challenges' with the subtitle 'Test your skills by engineering prompts to match target outputs'.

Four summary cards are shown: '12 Challenges Won', '18 Total Attempts', '67% Success Rate', and '#4 Global Rank'. Below these are tabs for 'Active Challenges' and 'Completed'.

The 'Active Challenges' section lists three challenges:

- JSON Object Generation** (+150 XP): 'Generate a prompt that produces a specific JSON structure'. Difficulty: Easy. Time: 10 min. Attempts: 234. Success: 68%. A 'Start Challenge' button is present.
- Code Refactoring Prompt** (+300 XP): 'Create a prompt that generates clean, refactored code'. Difficulty: Medium. Time: 20 min. Attempts: 156. Success: 45%. A 'Start Challenge' button is present.
- Multi-Step Reasoning** (+80 XP): 'Engineer a prompt that solves complex problems step-by-step'. Difficulty: Hard. Time: 30 min. Attempts: 12. Success: 32%. A 'Start Challenge' button is present.

The 'Leaderboard' section shows 'Top performers this month':

- Alice Chen: 6850 XP + 12 badges
- Bob Smith: 4720 XP + 11 badges
- Carol Davis: 6680 XP + 10 badges
- You (John Doe): 2450 XP + 5 badges
- Eve Wilson: 2380 XP + 5 badges

The 'Challenge Tips' section provides advice:

- Analyze the target output structure carefully
- Use clear, specific instructions
- Test your prompt in the playground first
- Pay attention to formatting details

The 'Rewards' section lists bonuses:

- Perfect Score (100): +50 XP Bonus
- First Attempt Win: +25 XP Bonus
- Speed Bonus (<5 min): +15 XP Bonus

The 'How Challenges Work' section explains the process:

Target Output:

```
{ "name": "John Doe", "role": "Engineer", "skills": ["Python", "JavaScript", "AI"] }
```

Your Task:

Write a prompt that generates the exact output above...

Goal: Create a prompt that produces the exact target output. The AI will compare your result with the target and score your accuracy.

Fig. 10: Challenges

This sub-view details the gamified "Reverse Engineering Challenges" page of the platform. It shows active performance metrics including challenges won, total attempts, overall success rate, and global ranking. The interface presents active prompt optimization tracks (e.g., JSON Object Generation, Code Refactoring, Multi-Step Reasoning) with varying difficulties. It also includes a live leaderboard showing rankings of participants, contextual challenge tips, and potential milestones rewards (such as speed bonuses or perfect score badges) to maximize engagement.

Marketplace

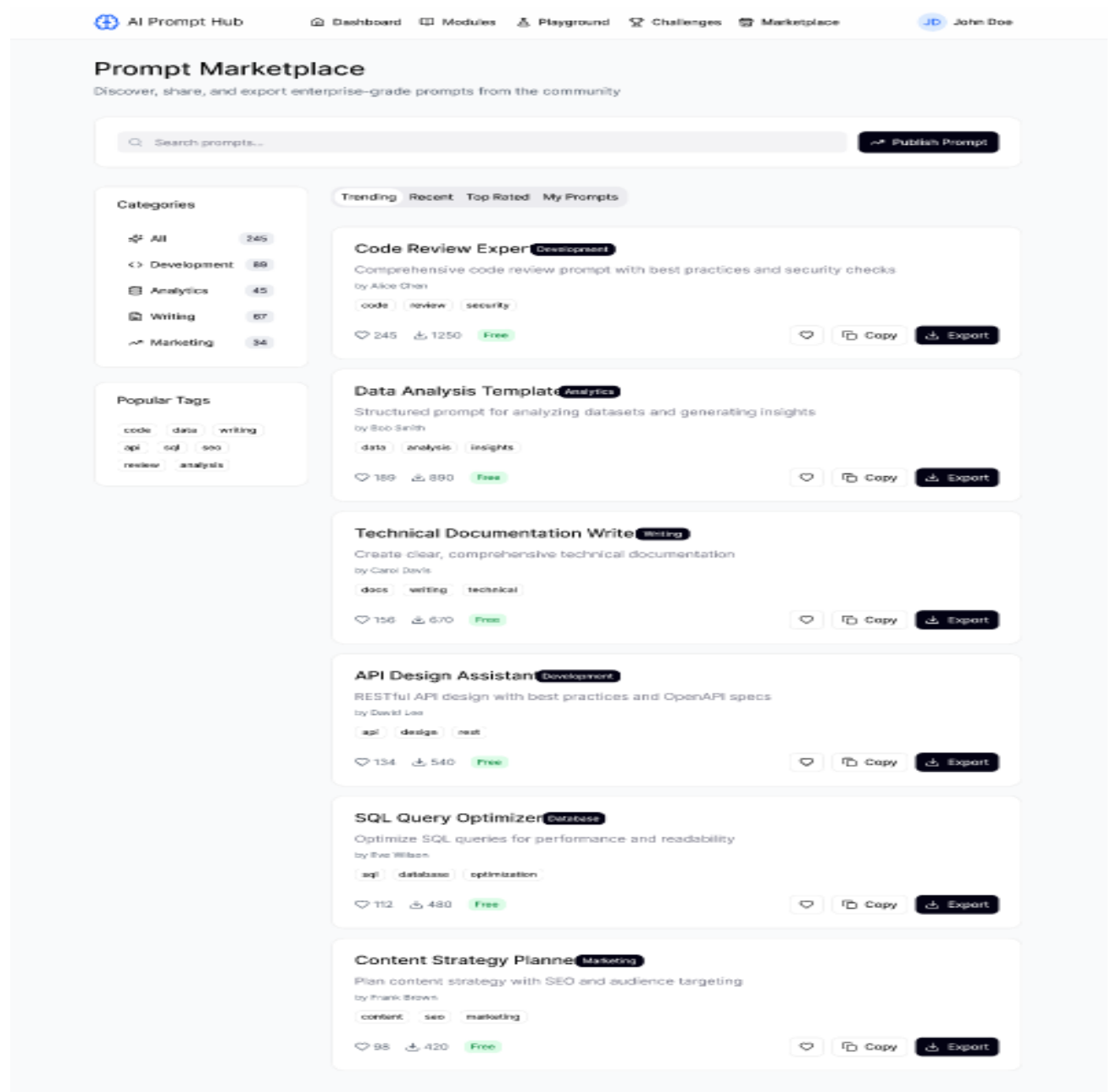


Fig. 11: Marketplace

This sub-view presents the "Prompt Marketplace" page, a collaborative environment where users can discover, share, and export enterprise-grade prompts from the community. It features a sidebar navigation categorized into distinct fields like Development, Analytics, Writing, and Marketing, along with a "Popular Tags" cluster. The central feed distributes prompt templates structured into multi-tab menus (Trending, Recent, Top Rated, and My Prompts). Each listing displays specific tags, engagement metrics (likes and downloads), pricing flags, and interactive quick actions to "Copy" or "Export" the prompts.

2.7 AI Interaction Page

The screenshot displays the "AI Prompt Playground" interface. At the top, a navigation bar includes "AI Prompt Hub", "Dashboard", "Modules", "Playground", "Challenges", and "Marketplace", along with a user profile "JD John Doe". The main heading is "AI Prompt Playground" with the subtitle "Test your prompts and get real-time AI evaluation and feedback".

The interface is divided into several sections:

- Write Your Prompt:** A text area for crafting prompts, with a "Clear" button. Below the text area are "Evaluate Prompt" and "Optimize" buttons.
- Evaluation Results:** A section showing the AI-powered analysis of the prompt. It includes a "Score" tab (selected), "Feedback", and "Optimized" tabs. The score is 84, displayed in a large blue circle. Below the score, it says "Good Quality Prompt" and "Your prompt shows strong structure but can be improved".
- Quick Templates:** A section with common prompt structures: "Role-Based Prompt" (Act as a [role]...), "Few-Shot Example" (Here are examples...), and "Chain-of-Thought" (Let's think step by step...).
- Evaluation History:** A table showing previous evaluations:

Score	Prompt	Action
92	Code Review	View
84	Data Analysis	View
78	Content Writing	View

Below the evaluation results, there are three metrics:

- Clarity:** 92 (Green checkmark icon)
- Specificity:** 78 (Blue line graph icon)
- Constraints:** 82 (Yellow warning icon)

Fig. 12: AI Prompt Playground

This page displays the interactive "AI Prompt Playground" where users can craft, test, and refine their custom prompts. It features a text editor to write prompts, a button to trigger real-time AI-powered analysis ("Evaluate Prompt"), and an optimization option. The system evaluates the prompt structure to return an exact quality score (e.g., 84/100) alongside deep breakdown scores for Clarity, Specificity, and Constraints. Quick templates, optimization suggestions, and a sidebar showing past evaluation history are also included to assist the user.

Prompt Feedback Analysis

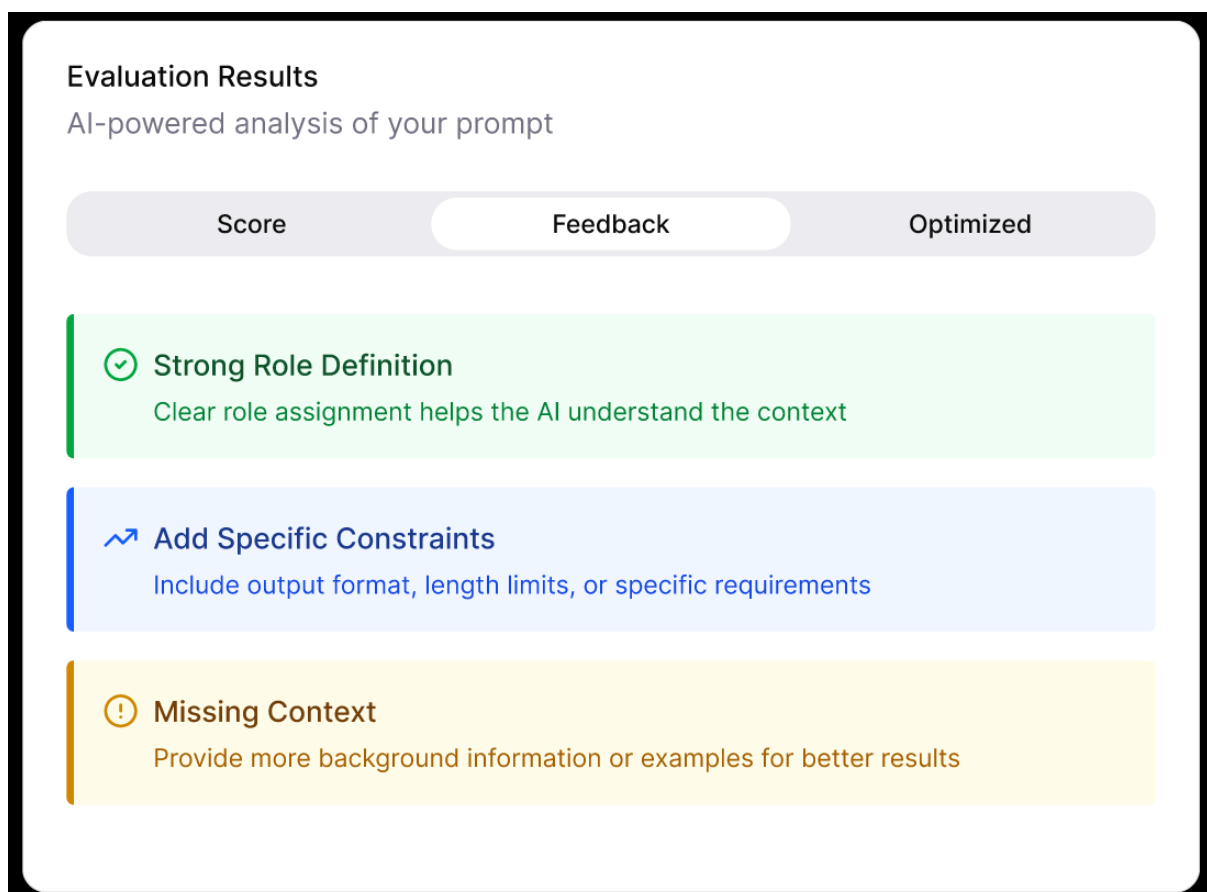


Fig. 13: Prompt Feedback Analysis

This sub-view highlights the detailed automated critique under the "Feedback" tab of the Evaluation Results interface. The system provides layered, color-coded diagnostic feedback on the user's prompt structure: confirming positives like "Strong Role Definition" (clear context assignment), alongside actionable improvement recommendations such as "Add Specific Constraints" (output

format, length limits) and flagging weaknesses like "Missing Context" to help users iteratively enhance their prompt engineering skills.

Optimized Prompt Output

The screenshot displays a user interface for AI prompt evaluation. At the top, the heading "Evaluation Results" is followed by the subtitle "AI-powered analysis of your prompt". Below this, there are three tabs: "Score", "Feedback", and "Optimized", with the "Optimized" tab being the active selection. The main content area, which has a dark blue background, contains a re-engineered prompt: "Act as a Senior Software Engineer with 10+ years of experience in code review. Review the following code segment and provide detailed feedback focusing on: 1. Code quality and best practices 2. Performance optimization opportunities 3. Security considerations 4. Maintainability improvements Format your response as: - Issue: [description] - Severity: Low/Medium/High - Suggestion: [specific improvement] Provide 3-5 actionable suggestions." At the bottom of the interface, there are two buttons: "Copy Optimized" (with a clipboard icon) and "Test This Prompt" (with a play icon).

Fig. 14: Optimized Prompt Output

This sub-view highlights the "Optimized" tab within the AI Evaluation Results section. When a user submits an unstructured prompt, the built-in system intelligently re-engineers it into a highly effective, structured format—incorporating role definitions, structural guidelines, and clear formatting rules (e.g., "Act as a Senior Software Engineer..."). Users can easily interact with this optimized version via quick action buttons to either "Copy Optimized" to their clipboard or instantly execute it using the "Test This Prompt" feature.

2.8 Admin Panel

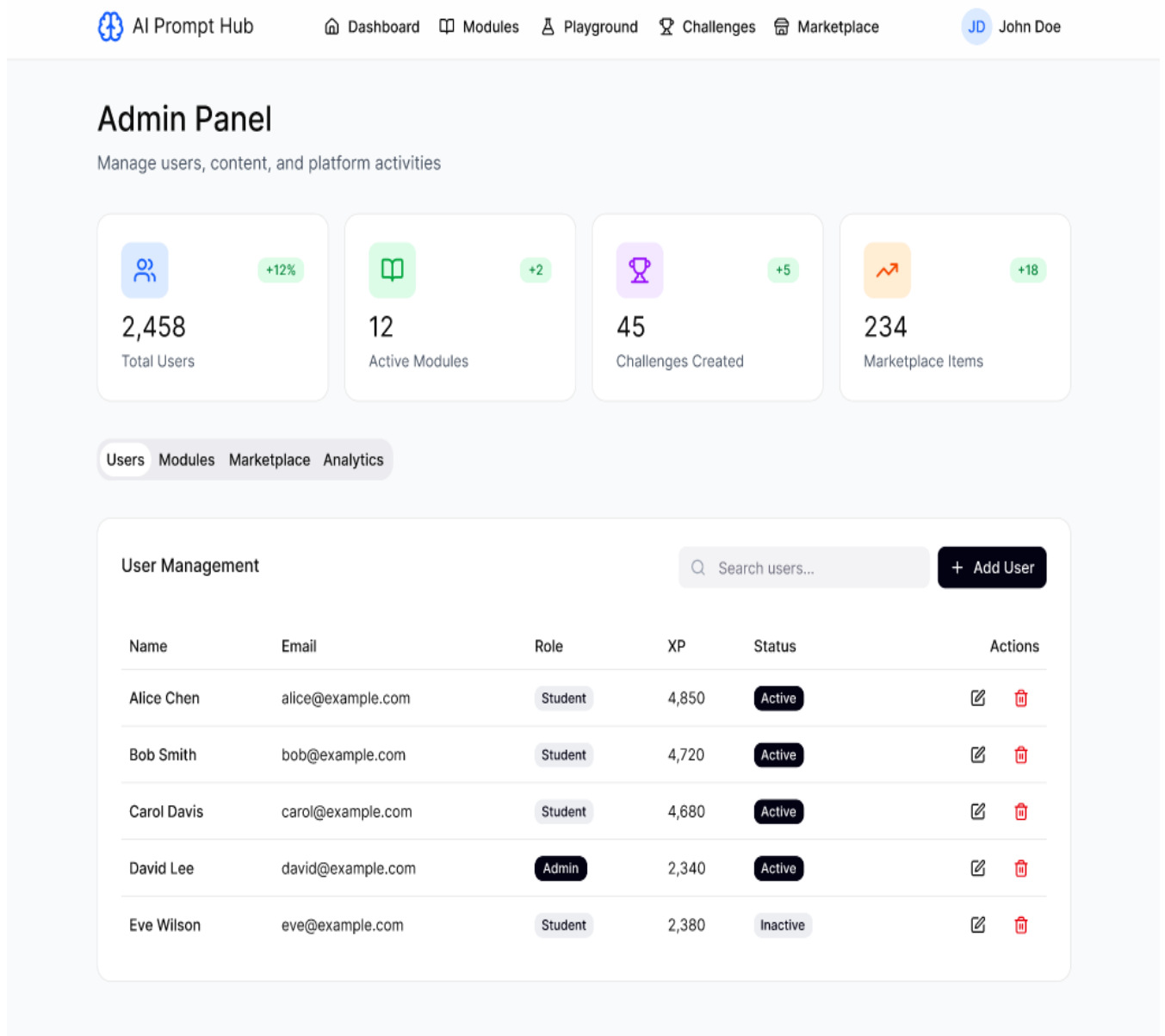
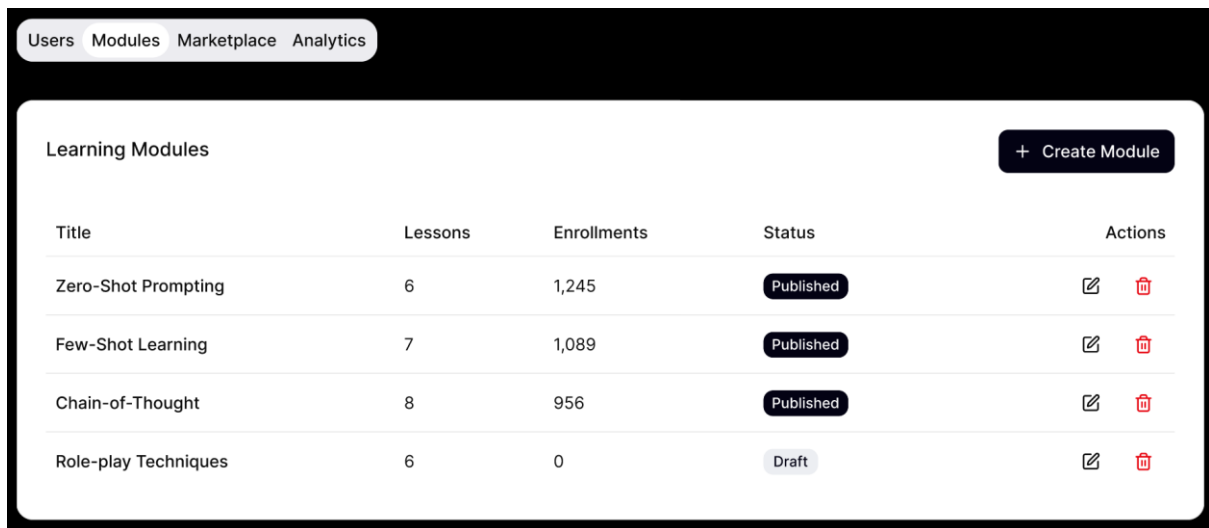


Fig. 15: Admin Panel

This page displays the administrative backend dashboard titled "Admin Panel," designed to manage users, content, and platform activities. The top section presents high-level system analytics cards representing Total Users (2,458), Active Modules (12), Challenges Created (45), and Marketplace Items (234). Below the metrics, a specialized "User Management" dashboard features navigation tabs (Users, Modules, Marketplace, Analytics), a comprehensive search bar, an "+ Add User" functionality, and an interactive database table to control, monitor, edit, or delete user records and roles.

Learning Modules Management

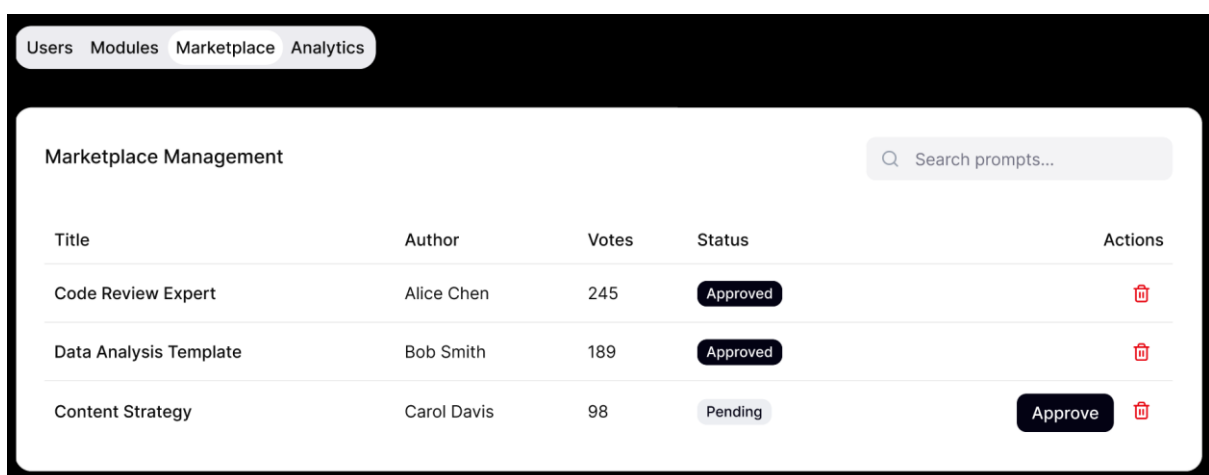


Learning Modules					+ Create Module
Title	Lessons	Enrollments	Status	Actions	
Zero-Shot Prompting	6	1,245	Published		
Few-Shot Learning	7	1,089	Published		
Chain-of-Thought	8	956	Published		
Role-play Techniques	6	0	Draft		

Fig. 16: Learning Modules Management

This sub-view illustrates the "Modules" management tab within the Admin Panel dashboard. It provides administrators with a structured data table to monitor course content across the platform. The dashboard tracks core content metrics including module titles (e.g., Zero-Shot Prompting, Few-Shot Learning), total lesson counts, live student enrollment stats, and publication states, enabling admins to manage existing syllabus streams or transition content from "Draft" status to "Published".

Marketplace Management



Marketplace Management					Search prompts...
Title	Author	Votes	Status	Actions	
Code Review Expert	Alice Chen	245	Approved		
Data Analysis Template	Bob Smith	189	Approved		
Content Strategy	Carol Davis	98	Pending	Approve	

Fig. 17: Marketplace Management

This sub-view highlights the "Marketplace Management" tab inside the Admin Panel. It gives administrators full moderation control over community-submitted

prompts on the platform. The data grid tracks key parameters including prompt titles (e.g., Code Review Expert, Data Analysis Template), author identity, community voting stats, and moderation states—allowing admins to review and transition user content from "Pending" review to "Approved" for public listing.

Platform Analytics

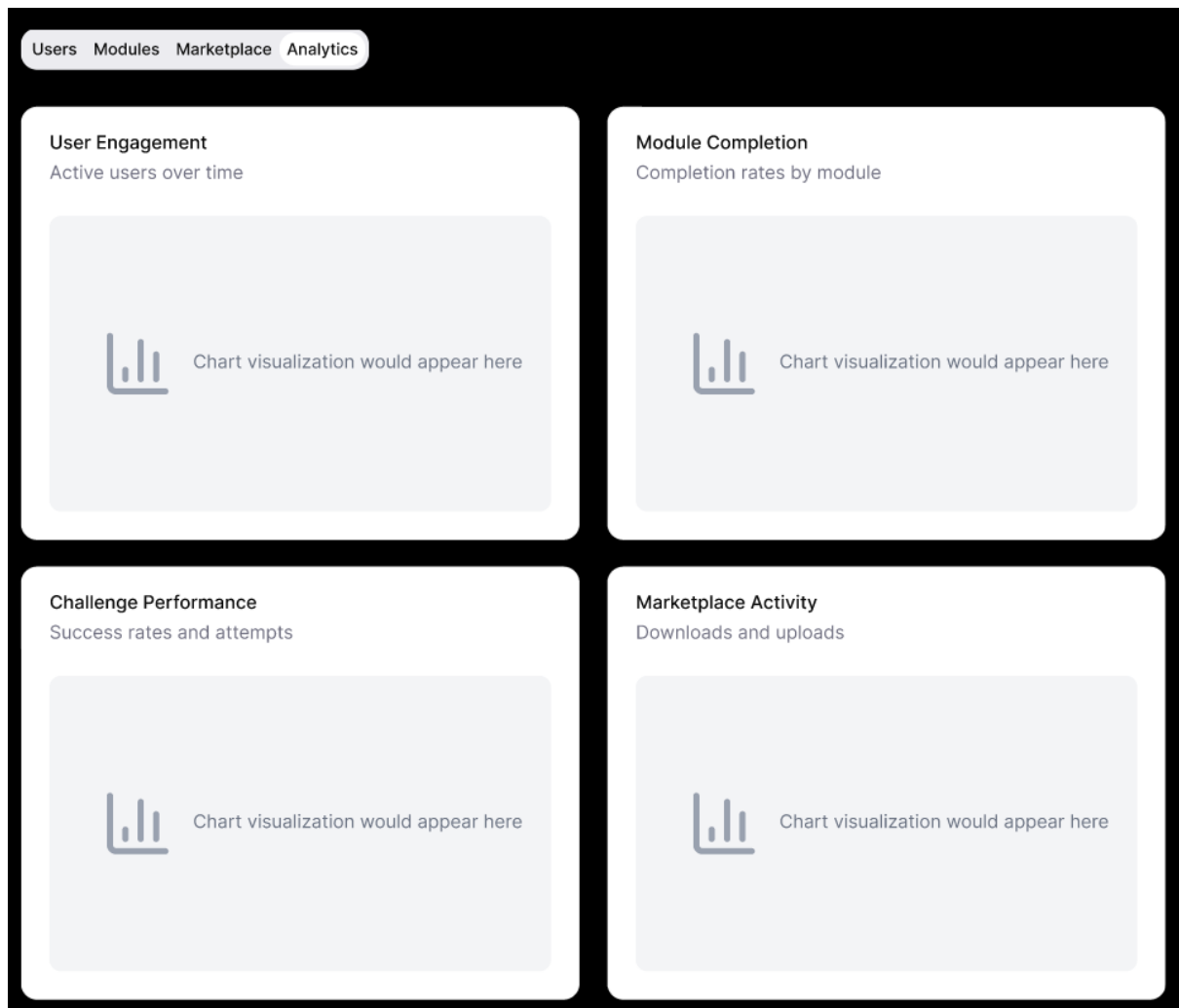


Fig. 18: Platform Analytics

This sub-view presents the "Analytics" tab inside the Admin Panel dashboard. It contains structured containers for data visualization and chart generation to track overall system behavior. The layout tracks crucial administrative insights over time, including "User Engagement" (active user tracking), "Module Completion" rates, "Challenge Performance" (success metrics and submissions), and "Marketplace Activity" (download and upload logs).

3. User Flow Diagram

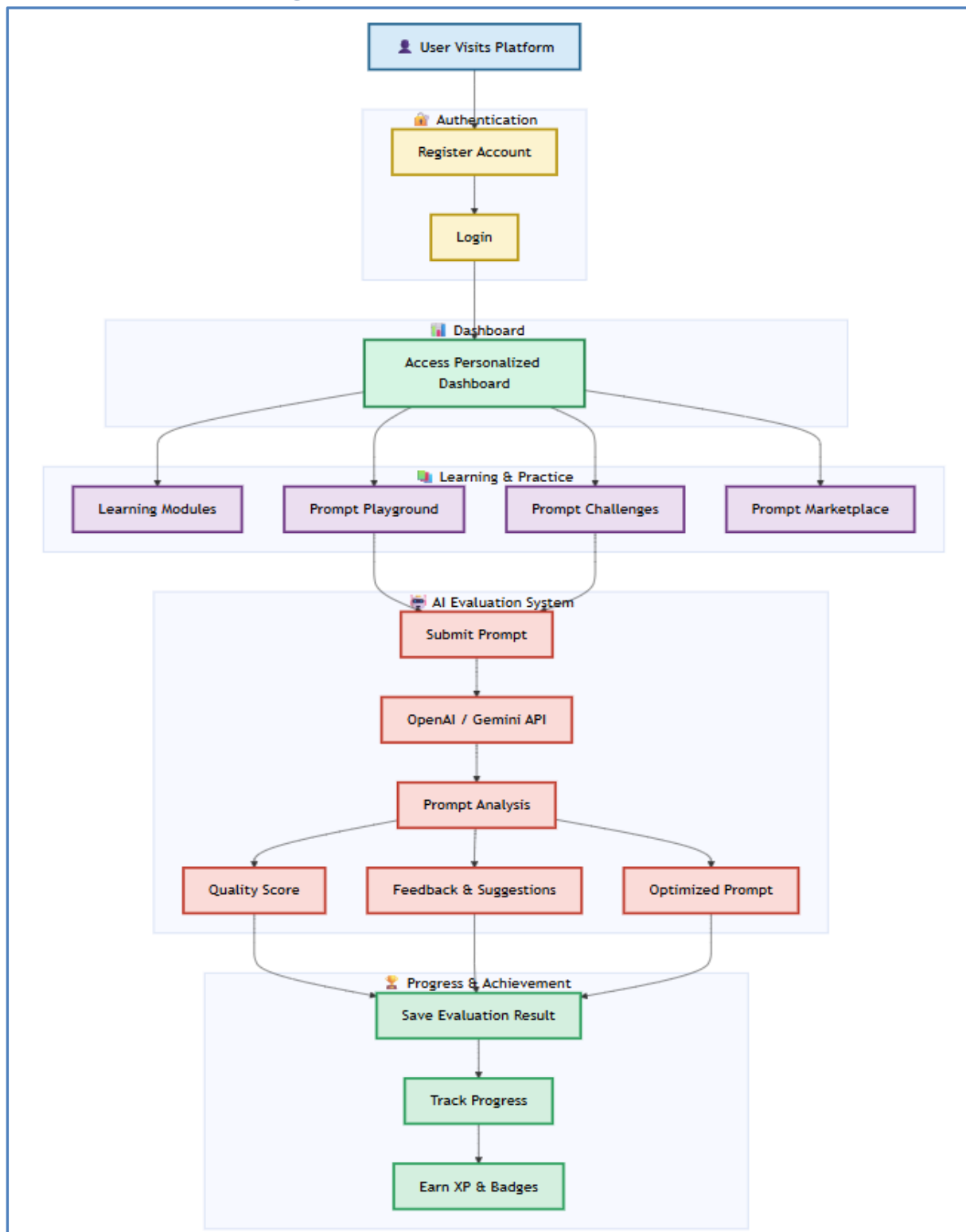


Fig. 19: User Flow Diagram

4. Development Roadmap

The development roadmap for the AI Prompt Engineering Learning Hub is based on the Software Development III course workflow. The roadmap outlines the planned development phases and expected outcomes for the remaining weeks of the semester.

Week	Development Phase	Expected Outcome
Week 06	Backend Development	Implementation of authentication system, database integration, and backend APIs for core system functionalities.
Week 07	Frontend Development	Development of user interfaces including authentication pages, dashboard, navigation system, and AI interaction screens.
Week 08	AI Integration	Integration of Google Gemini API for prompt evaluation, prompt optimization, quality scoring, and AI-generated feedback.
Week 09	Feature Completion	Completion of all major project features, user experience improvements, performance optimization, and bug fixes.
Week 10	Testing and Debugging	Functional testing, API testing, UI testing, AI response testing, bug identification, and issue resolution.
Week 11	Deployment	Deployment of frontend and backend systems to cloud platforms and preparation of a live project environment.
Week 12	Documentation	Preparation of final report, user manual, API documentation, and project documentation updates.

5. Team Task Distribution

To ensure efficient project development, responsibilities have been assigned according to each member's expertise and designated role.

Team Member	Student ID	Assigned Responsibilities
Rahmin Hossain	11230121156	Team Leadership, Database Design, Database Integration, Development Planning, Documentation Review
Lamiha Faija	11230121140	UI/UX Design, Frontend Development, User Interface Research and Implementation
Masum Billah	11230121149	Backend Development, API Development, System Testing, Quality Assurance
Fatema Akter Elma	11230121142	AI Integration, Prompt Evaluation System, AI Feature Research and Implementation

6. Final Technology Stack

The AI Prompt Engineering Learning Hub will be developed using the following technology stack.

➤ Frontend Technologies

- React.js
- Tailwind CSS
- Redux Toolkit

Purpose: To create a modern, responsive, and interactive user interface that provides a seamless learning experience.

➤ Backend Technologies

- Node.js
- Express.js
- JWT Authentication
- Google OAuth

Purpose: To handle application logic, user authentication, API development, and communication between the frontend, database, and AI services.

➤ Database

- MongoDB

Purpose: To store user information, learning progress, badges, challenge records, prompt history, and marketplace content.

➤ AI Services

- Google Gemini API

Purpose: To provide prompt evaluation, prompt optimization suggestions, quality scoring, semantic feedback generation, and AI-assisted learning support.

➤ Development Tools

- Git
- GitHub
- Figma
- Draw.io

Purpose: To support version control, team collaboration, UI/UX design, and system diagram creation.

➤ Deployment Platforms

- Vercel (Frontend Deployment)
- Render (Backend Deployment)

Purpose: To host and deploy the application components in a scalable and accessible environment.

7. Conclusion

This document presents the UI/UX design, user flow, development roadmap, team responsibilities, and technology stack of the proposed system. These designs and plans will serve as the blueprint for the implementation phase and help ensure a smooth and efficient development process.